INCH-POUND

A-A-50632
10 February 1995

COMMERCIAL ITEM DESCRIPTION

STRAINER, SIMPLEX, CONTINUOUS SELF-CLEANING

The General Service Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description covers continuous self-cleaning simplex strainers.

1.2 Intended use. The strainers specified are intended for installation on the discharge side of the pump in the fuel/ lubricating oil piping systems.

2. CLASSIFICATION

2.1 Classification. Units should be of the following types and classes as specified (see 7.1):

Type I - 150 pound per square inch gage (psig) pressure rating
Type II - 300 psig pressure rating
Type III - 600 psig pressure rating

Class 1 - 75 Micron rating
Class 2 - 40 Micron rating
Class 3 - 25 Micron rating

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03R42, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160.

FSC 4730

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3. SALIENT CHARACTERISTICS

3.1 Codes and standards. Strainer shall comply with the applicable requirements of the ASME Boiler and Pressure Vessel Code, Section VIII and IX.

3.2 Design. Each self-cleaning simplex type strainer shall remove particulate contaminants from fuel or lube oil. The self-cleaning action shall be continuous, and shall have the ability to periodical-ly backwash the strained fluid through the strainer element in segments such that the particulate contaminants are flushed free of the strainer element and down into a large sump. The self-cleaning action of each unit shall be driven by a hydraulic motor which uses strained system fluid as its sole source of power.

3.3 General construction. Each strainer assembly shall consist of a housing, strainer element(s) and self-cleaning mechanism(s). The cross-sectional flow area throughout the unit shall be equal to or greater than that of the piping to which the strainer is connected.

3.3.1 Housing. The strainer housing, cover, flanges, and other items which form the pressure boundary shall conform to the require-ments of the ASME Boiler and Pressure Vessel Code, Section VIII.

3.3.2 Cover. Cover shall be secured to the housing by nuts and bolts. The cover shall effect a positive seal on a gasket or O-ring recessed in the body. Removal of the cover shall provide access to all internal components and shall not require unbolting the inlet and outlet strainer piping connections. Cover lifting devices with integral supports shall be required for covers weighing more than 30 pounds.

3.3.2.1 Anti-spray feature. The strainer cover shall contain a device to deflect fluid spray downward in the event of sealing device failure. The spray deflector shall remain in position at all times while the strainer is pressurized.

3.3.2.2 Inter-lock feature. The strainer cover shall contain a locking device to prevent disassembly while the unit is pressurized.

3.3.3 Piping connections. Piping connections for 3.3.3.1, 3.3.3.2 and 3.3.3.3 shall be either welded, flanged, straight threaded unions, or boss connections to the strainer housing. Straight threaded unions, or boss connections shall be in accordance with ANSI J 1926. Tapered threaded connections shall not be allowed.

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3.3.3.1 Inlet and outlet connections. Inlet and outlet connections of each strainer shall be the same type (see 3.3.3) and size as specified (see 7.1). Unless otherwise specified (see 7.1), both inlet and outlet connections shall be flanged. Flanges shall be sized and drilled to conform with ANSI B16.5, unless otherwise specified (see 7.1).

3.3.3.2 Gage valve connections. When specified (see 7.1), gage valve connections shall be included at both the inlet and outlet. Straight threaded unions, or boss connections shall be in accordance with ANSI J 1926.

3.3.3.3 Vent connections. Vent valves shall be installed with the underside of the seat subject to system pressure.

3.3.3.4 Drain connections. Two drain connections not less than 3/4 inch 16 UNF-2B straight threads shall be provided in accordance with ANSI J 1926, one at a low point in the contaminated oil compartment and one at the low point in the clean oil compartment. These connec-tions shall be furnished with caps or plugs.

3.3.3.4.1 Sludge cleanout. A flanged sludge cleanout connection shall be positioned 90° or 180° to the inlet/outlet. This connection shall provide an access area of not less than 6 square inches. This connection shall be furnished with a blind flange.

3.3.4 Strainer element construction. Strainer element shall be screen type construction, consisting of a supporting framework (strength member) with a wire screen as the straining medium. The supporting framework (strength member) shall be plate construction. The strainer element shall be made of corrosion-resistant steel. Wire screen edges and seams shall be folded back upon themselves and welded or brazed to preclude fraying. All seam welds shall be continuous. The strainer element shall be held securely in place when the strainer is assembled.

3.3.4.1 Supporting element. The supporting element shall be designed to facilitate easy removal of the filter element, and element support as an assembly. The supporting element must satisfy the flow area requirements of section 3.4.

3.3.4.2 Straining element. Unless otherwise specified (see 7.1), the straining element shall be attached to the supporting element in such a manner that it can easily be removed and replaced from the element support. The free flow area through the assembly of the strainer element, and element support shall be not less than 2 times the cross-sectional area of the strainer discharge

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connection. The element micron rating shall be specified by 2.1. Unless other wise specified (see 7.1), the minimum beta ratio shall be 75.

3.3.5 Self-cleaning mechanism. The self-cleaning action of each unit shall be operated by a hydraulic motor included as a part of the strainer. Each hydraulic motor shall use strained system fluid as its sole source of power. The fluid consumption of the motor(s) shall not exceed 4 percent of the total flow through the strainer assembly.

3.3.5.1 Backflush structure. The backflush structure shall periodically clean the entire strainer element by backflushing system fluid through the straining element. The structure shall clean the strainer element in segments, so as not to disrupt the system flow through the strainer at any time. All impurities separated shall be deposited within a sludge compartment isolated from the strained liquid.

3.3.6 Sludge compartment. Unless otherwise specified (see 7.1), the sludge compartment shall have a volume of not less than 320 cubic inches for each unit.

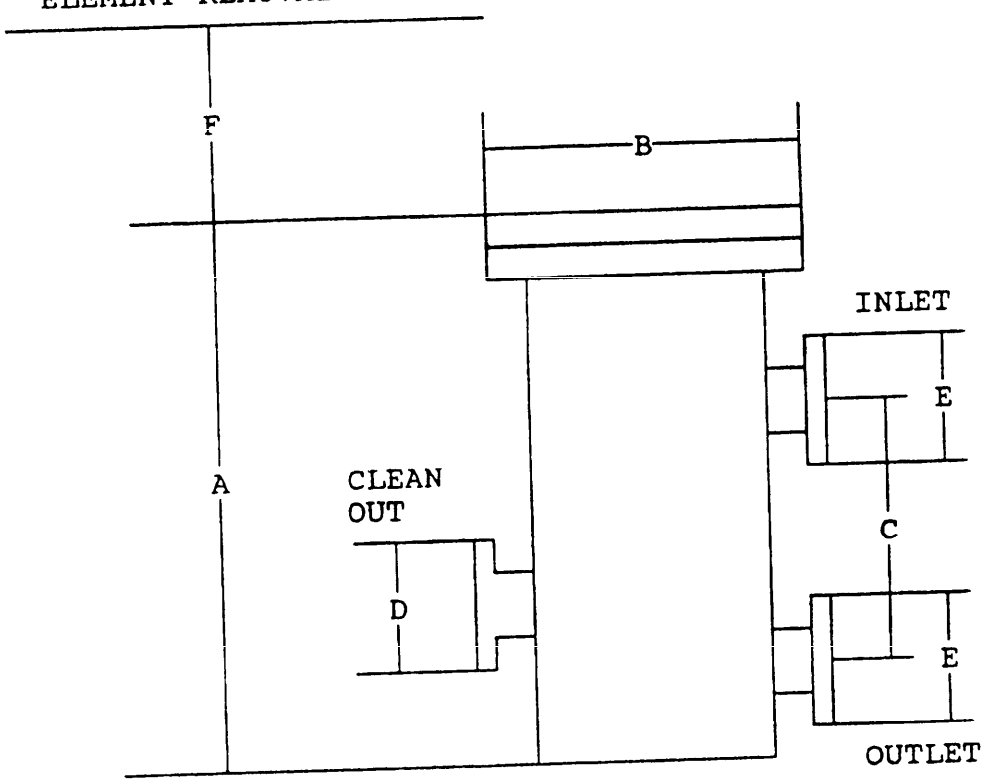
3.3.7 Material. Unless otherwise specified (see 7.1), materials shall be as specified in table I.

TABLE I - Materials for strainers.

Part	Material	Applicable Documents	Remarks
Bodies, covers	CRES	ASTM A 276	S31600
	CRES Plate	ASTM A 167	S30400
Self-cleaning mechanism	CRES Plate	ASTM A 167	S30400
	CRES	ASTM A 276	
Threaded fasteners	Stainless Steel	ASTM F 593	S30600, S31600
		ASTM F 594	
"O" rings	Fluorocarbon	MIL-R-83248/1	
Strainer element	CRES Plate	ASTM A 167	S30400
	CRES Wire	ASTM E 437	S30400, S31600

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3.3.8 Envelope dimensions. Unless otherwise specified (see 7.1), dimensions shall be as specified on figure 1.



Rated flow (gpm)	75	150	300	450	600	750	900	1050	1200
A	51	51	51	53	54	54	55	55	56
B	10-7/8	10-7/8	27-1/2	27-1/2	32	32	36	36	42
C	30	30	30	30	32	32	36	36	36
D	4	4	5	6	8	8	10	10	10
E	2	3	4	6	6	6	8	8	8
F	25	25	25	25	25	25	25	25	25

FIGURE 1. Envelope dimensions.

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NOTE: All dimensions are in inches.

Flange drilling, straddles center line.

Unless otherwise specified the motor exhaust flange shall be 1-1/2".

Unless otherwise noted all dimensions ± 1/4".

Foundation base shall be designed to accommodate 4 mounting bolts, 1" diameter, 90 degree apart, on a circle of diameter "B".

3.4 Performance. Strainers shall be capable of operating in accordance with all requirements of this specification when subjected to the motion and attitude constraints of MIL-STD-1399, Section 301.

3.4.1 Capacity. The design flow rate for a single element housing shall be 150 gpm for fluids with viscosities less than or equal to 500 Seconds Saybolt Universal (SSU). Additional capacity can be achieved by specifying multiple elements within a single housing (see 7.1).

3.4.2 Pressure. Unless otherwise specified (see 7.1), the strainer shall be suitable for operation in the range of 60 psig to design pressure (see 2.1).

3.4.3 Temperature. Unless otherwise specified (see 7.1), the strainer shall be suitable for operation at a design temperature not greater than 250°F.

3.4.4 Self-cleaning rate. The pressure drop through the strainer assembly shall not be greater than 3 psi, unless otherwise specified (see 7.1), during operation. Specifically, the self-cleaning mechanism shall be capable of maintaining the pressure drop below this value when contaminated fluid is being pumped through the strainer at the design flow specified in 3.4.1.

3.5 Identification. Identification plates shall be provided in accordance with MIL-P-15024 and MIL-P-15024/5, and shall include the following information:

a. Commercial Item Description (CID) number.
b. Type and class.
c. Manufacturer.
d. Contract Number.
e. The nominal pipe size of the inlet and outlet connections.

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f. Design pressure.

3.5.1 Part numbering of interchangeable parts. The manufacturer's part number and drawing number shall be the same. All fasteners shall be identified by an industry part number which identifies both size and material.

3.6 Mechanical shock. The strainer assembly, including the differ-ential pressure gauges, shall meet the high impact shock requirements of MIL-S-901 for grade A, class 1 equipment.

3.7 Vibration. The self-cleaning simplex strainers shall be designed and tested to function satisfactorily under the requirements of MIL-STD-167-1 Type 1 environmental vibration.

3.8 Acoustic requirement. When operating, the strainer assembly shall conform to the airborne noise requirements of MIL-STD-740 for grade C equipment.

3.9 Welding and allied processes. Fabrication welding and inspec-tion, and casting inspection and repair, shall be in accordance with MIL-STD-278, class P-1. Welding for pressure boundary components shall be in accordance with the ASME Boiler and Pressure Vessel Code, Section IX.

3.10 Workmanship. The strainer and its components shall be free from blow holes, porosity, hard spots, shrinkage defects and cracks. All surfaces shall be smooth and clean. Where dimensions and tolerances affect interchangeability, operation, or performance, they shall be held.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

4.2 Prohibited material. Zinc or cadmium plating shall not be used on any part of the strainer.

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4.3 Mercury exclusion. The strainer shall be free of mercury contamination. During the manufacturing process, tests and examinations (see 7.3), the product to be offered for acceptance shall not come in direct contact with mercury or any of its components, nor with any mercury containing device employing a single boundary of containment.

5. QUALITY ASSURANCE PROVISIONS

5.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of the inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

5.1.1 Responsibility for compliance. All items must meet the requirements of sections 3 and 6. The inspections set forth in this specification shall become a part of the contractor's inspection system or quality program. The absence of inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that products or supplies submitted to the Government for acceptance comply with requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of the defective material, nor does it commit the Government to acceptance of defective material.

5.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 5.3).
b. Quality conformance inspection (see 5.4).

5.3 First article inspection. First article inspection shall consist of one strainer of each size, type, class, and identical design undergoing the examination and tests as specified in table II.

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TABLE II. First article and quality conformance inspection.

Examination and Tests	First Article	Quality Conformance
Examination	5.6	5.6
Shock	5.7.4.1	---
Vibration	5.7.4.2	---
Hydrostatic	5.7.1	5.7.1
Flow test	5.7.2	---
Hydraulic motor flow	5.7.2.5	5.7.2.5
Acoustic requirement	5.7.4.3	---
Capacity	5.7.2.3	---
Pressure range	5.7.2.4	---
Inclined operation	5.7.2.8	---
Self-cleaning rate	5.7.3	5.7.3
Strength of internals	5.7.2.7	---
Straining element efficiency	5.7.2.6	---
Spray deflector	5.7.5	---

5.4 Quality conformance inspection. All assembled strainers shall undergo quality conformance testing as specified in table II. Visual and dimensional examination shall consist of selecting strainer(s) in accordance with 5.5 and inspecting them in accordance with 5.6.

5.5 Sampling for visual and dimensional examination.

5.5.1 Lot. For sampling, a lot shall consist of strainers of the same type, class, size, and identical design offered for delivery at one time.

5.5.2 Sampling. As a minimum, the contractor shall select a sample quantity of strainers in accordance with table III and inspect them in accordance with 5.6. If one or more defects are found in any sample, the entire lot shall be rejected. The contractor has the option of screening 100 percent of the rejected lot for the defective characteristic(s) of providing a new lot

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which shall be inspected in accordance with the sampling plan.

Table III - Sampling for visual and dimensional examination.

Lot Size Number of Strainers	Sample Size
5 and under	All
6 to 8	5
9 to 15	7
16 to 25	10
26 to 40	15
41 to 65	25
66 to 110	35
111 to 180	50
181 to 300	75

5.6 Visual and dimensional inspection. The visual and dimensional inspection shall consist of the inspections listed in table IV. Any strainers having one or more defects shall be rejected.

TABLE IV. Visual and dimensional inspection.

Inspection	Requirement paragraph
Presence of ASME stamp	3.3.1
Presence of cover lifting device, if required	3.3.2
Presence of anti-spray feature integral with cover	3.3.2.1
Presence, sizes, and locations of connections	3.3.3.1
Presence, completeness, legibility, and correctness of identification	3.5
Quality of workmanship	3.10
Vent valve connection	3.3.3.3
Drain valve connection	3.3.3.4
O-rings	3.3.2

5.7 Tests.

5.7.1 Hydrostatic pressure test for strength and porosity. The test shall be an internal pressure test with water or system fluid. Each assembled strainer shall be tested to a pressure 150 percent of the design pressure for a duration of 60 minutes. Any strainers not maintaining zero leakage shall be rejected.

5.7.2 Flow Tests. Unless otherwise specified (see 7.1), the test fluid viscosity shall be within ten percent of 500 Seconds Saybolt Universal (SSU). Within this range, test results may be analytically corrected for the deviation of the viscosity from the specified value. The temperature of the fluid must be less than

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the maximum temperature specified in 3.4.3.

5.7.2.1 Test setup. For the fluid flow tests listed in 5.7.2, the strainer assembly shall be installed in a test loop in accordance with ISA-S75.02. The facility shall include instrumentation to measure the flow through the strainer assembly, the total flow over a period of time through the hydraulic motor(s), the inlet pressure to the strainer assembly, the pressure drop across the strainer, and the temperature at the inlet of the strainer assembly. Instrumenta-tion used in these tests shall be of laboratory precision type and shall be calibrated at intervals properly spaced to insure laboratory accuracy. A visual flow indicator shall be installed in the exhaust line from each hydraulic motor in the strainer assembly. Each flow indicator shall show hydraulic motor flow.

5.7.2.2 Conditions. The requirements of this paragraph shall apply to the capacity test, the pressure range test, and the hydraulic motor flow test. The test flow rate shall be within ten percent of that specified in 3.4.1. The flow rate shall be the product of multiplying the number of elements in the strainer assembly by 150 gpm, plus or minus ten percent. Within this range, test results may be corrected analytically for the deviation of the flow rate from the specified value. The hydraulic motor exhaust pressure downstream of any required orifice shall be the ambient atmospheric pressure. Each of the tests covered by this paragraph shall be performed at two different inlet pressures to the strainer assembly, which shall be the following:

a. A pressure within ten psi of the minimum pressure specified in 3.4.2, but not less than this minimum pressure.
b. A pressure within ten psi of the maximum pressure specified in 3.4.2, but not greater than this maximum pressure.

5.7.2.3 Capacity test. This test shall be performed in accordance with 5.7.2.2. The pressure drop across the strainer assembly shall not exceed 3 psi, for a fluid with a viscosity of 500 SSU.

5.7.2.4 Pressure range test. This test shall be performed in accordance with 5.7.2.2. The flow indicator(s) shall be observed. Proper hydraulic motor operation is shown by the pulsating motor flow. The lack of pulsating flow, even if a continuous flow is being indicated, shall constitute failure of this test.

5.7.2.5 Hydraulic motor flow test. This test shall be performed in accordance with 5.7.2.2. At each inlet connection pressure and flow shall be maintained for fifteen minutes, during which time the flow through the hydraulic motor(s) shall be totalled. Particular attention shall be paid to maintaining a constant flow rate through

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the strainer assembly. The average flow rate through the hydraulic motor(s) for this time shall be calculated. The fluid consumption of the hydraulic motor(s) shall conform to 3.3.5.

5.7.2.6 Straining element efficiency test.

5.7.2.6.1 Straining ratio (Strainer element efficiency). After completion of 5.7.2.7, the elements shall be subjected to a strainer element efficiency test using the specified test dust (see 5.7.3.2) and the test procedure specified in ANSI B93.31.

5.7.2.6.2 Retention test. The straining assembly and its straining elements shall be tested for effectiveness of straining using clean fluid of the viscosity specified in 5.7.2. Oil shall be pumped through the strainer at rated flow. At the rate of 10 grams per 4 minutes, 1.0 gram per gpm of test dust as specified in 5.7.3.2 shall be added. After all dust has circulated, the pressure differential through the strainer and total weight of test dust retained by the basket assembly shall comply with 3.3.4.2.

5.7.2.7 Strainer internals strength test.

5.7.2.7.1 Compression test. A differential pressure (outside-inside) shall be applied across the support/straining element assembly equal to 100 percent of the specified strainer design pressure. This differential may be accomplished by installing a strippable film on the straining element exterior. Then increasing the pressure to the maximum pressure drop is achieved and held for 20 minutes. Alternate methods of obtaining differential pressure will be considered by the command or agency concerned. Supporting or straining elements which distort, stretch or exhibit other weakness shall be rejected.

5.7.2.7.2 Expansion test. A differential pressure (inside-outside) shall be applied across the straining element assembly equal to 100 percent of the specified strainer design pressure. This may by accomplished by installing a strippable film on the straining element interior. Then increasing the pressure to the maximum pressure drop is achieved and held for 5 minutes. Alternate methods of obtaining differential pressure will be considered by the command or agency concerned. Straining elements which distort, stretch or exhibit other weaknesses shall be rejected.

5.7.2.8 Inclined operation test. The strainer assembly shall be inclined at an angle from the normal equal to the combination of the maximum permanent list and trim as specified (see 3.4). The capacity test (5.7.2.3), pressure range test (5.7.2.4), and hydraulic motor flow test (5.7.2.5) shall be repeated at this

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inclined position. The criteria for acceptance shall be as specified in each test paragraph.

5.7.3 Self-cleaning rate test.

5.7.3.1 Test fluid. Unless otherwise specified (see 7.1), the test fluid shall be 500 SSU.

5.7.3.2 Test contaminant. The test dust contaminant shall be of the proposed ISO test dust test grade identified table V. The dust is available from Powder Technology Inc., (PTI), P.O. Box 1464, Burnsville, MN 55337.

Table V - Test dust contaminant.

Micrometres size	Volume % content of test dust larger than	
	Coarse	Medium
5	88 - 89	79 - 82
10	77 - 79	54 - 58
20	61 - 64	29 - 33
40	38 - 41	11 - 13
80	10 - 12	0 - 1
Straining rating (micrometres)	Test dust grade	
10 - 30	Medium	
30 - 80	Coarse	

5.7.3.3 Test setup. The strainer assembly shall be installed in a test loop suitably equipped to pump the test fluid at the required pressures and flows. The facility shall include instrumentation to measure the flow through the strainer assembly, the inlet pressure to the strainer assembly, and the pressure drop across the strainer. All contaminated fluid necessary for this test shall be contained in a tank, from which it can be pumped through the strainer. The tank shall contain sufficient fluid to run this test at the flow specified in 5.7.3.4. Both the strained fluid and contaminant sludge shall be continuously returned to the tank.

5.7.3.4 Conditions. The test flow rate shall be within ten percent as specified in 3.3.2.1, that is, the flow rate shall be the product of multiplying the number of elements in the strainer assembly by 150 gpm, plus or minus ten percent. Within this range, test results may be corrected analytically for the deviation of the flow rate from the specified value. The hydraulic motor exhaust

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pressure downstream of any required orifice shall be the ambient atmospheric pressure. The pressure shall be within the range specified in 3.4.2 and the temperature shall be 150°F ± 25°F.

5.7.3.5 Procedure. The thoroughly mixed contaminated test fluid shall be pumped through the strainer assembly until the pressure drop across the strainer is increased to a minimum of 3 psi. The changing differential pressure across the strainer shall be recorded during the test. The concentration of contaminant may be increased up to three times in order to attain 3 psi within fifteen (15) days.

5.7.4 Mechanical tests.

5.7.4.1 Mechanical shock test. This test shall be conducted in accordance with MIL-S-901. All functions of the strainer assembly shall be unaffected by this test. Any unit not meeting MIL-S-901 requirements shall be rejected.

5.7.4.2 Vibration test. This test shall be conducted in accordance with MIL-STD-167-1. All functions of the strainer assembly shall be unaffected by this test. Any unit not meeting MIL-STD-167-1 requirements shall be rejected.

5.7.4.3 Acoustic requirement test. This test shall be conducted in accordance with MIL-STD-740. The sound energy limits imposed by MIL-STD-740 for grade C equipment shall not be exceeded.

5.7.5 Spray deflector effectiveness test. One strainer of each type, class, size and specific design shall be subjected to this test. The strainer shall be centered in drip pan 3-inches deep and of a size and shape to allow 4-inches clearance around the periphery. The strainer cover shall be installed with an O-ring which has been notched to cause leakage. The strainer shall be pressurized to maximum design pressure. This test shall be conducted 8 times with the O-ring notch moved clockwise 45 degrees between tests. Failure to pass this test in any leakage position shall require redesign of the deflector and retest.

5.8 Additional inspection. Where other specifications form a part of this specification, sampling, examination and tests shall be conducted as required by the pertinent specification, unless otherwise specified in the contract or order (see 7.1).

5.9 Inspection of preparation for delivery. Sample packages and the inspection of the packaging-preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 6 and the documents specified therein.

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6. PACKAGING

6.1 Commercial. Commercial packaging shall be in accordance with ASTM D 3951.

6.1.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two, 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage and storage.

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1 Acquisition requirements.

(a) Title, number, and date of this commercial item description.
(b) Type and class required (see 1.2)
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specified issue of individual documents referenced (7.2.1.1 and 7.2.2).
(d) Type and size of inlet and outlet connections and whether flanged or not flanged (see 3.3.3.1).
(e) Whether gage connections are required at inlet and outlet (see 3.3.3.2).
(f) The required design flow capacity, design pressure, range of fluid temperature and viscosity, and maximum allowable pressure drop (see 3.4.1, 3.4.2, 3.4.3, 3.4.4, 5.7.2, 5.7.3.1).
(g) Alternate strainer element design and characteristics (see 3.3.4.2).
(h) Alternate or additional design characteristics (see 3.3.6, 3.3.7, 3.3.8).
(i) If other specification are not to form a part of the specification for sampling, examination and tests (see 5.8).
(j) Special requirements or instructions (see 5.8).
(k) Quantity of technical manuals and drawings required (see 7.3).
(l) Additional training aids required (see 7.3.4).
(m) Whether repair parts are required (see 7.6.1).

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7.2 Sources of documents.

7.2.1 Government documents. The following documents of the issue in effect on date of invitation for bids of request for proposal, form a part of the specification to the extent specified.

7.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 7.1).

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High Impact): Shipboard Machinery, Equipment and Systems, Requirements For

MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment

MIL-P-15024/5 - Plates, Identification

MIL-R-83248 - Rubber Fluorocarbon Elastomer, High Performance Fluid, and Compression Set Resistant.

MIL-R-83248/1 - Rubber Fluorocarbon Elastomer, High Temperature Fluid, and Compression Set Resistant (O-rings, Class 1, 75 Hardness).

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II Internally Induced)

MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of The United States Navy

MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment

MIL-STD-1399 - Interface Standard for Shipboard Systems Section 301 Ship Motion and Attitude

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

7.2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicita-tion. Unless otherwise specified, the issues of the documents cited in the solicitation (see 7.1).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B 16.5 Steel Pipe Flanges and Flanged Fittings (DoD adopted)
B 93.31 Multi-Pass Method for Evaluating the Filtration Performance of a Fine Hydraulic Fluid Power Filter Element.
J 1926 Specification for Straight Thread O-ring Boss Port.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code - Section VIII - Pressure Vessels.
Boiler and Pressure Vessel Code - Section IX - Welding & Brazing Qualifications

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip. (DoD adopted)
A 276 - Standard Specification for Stainless and Heat-Resisting Steel Bars and Shapes. (DoD adopted)
D 3951 - Standard Practice for Commercial Packaging
E 437 - Standard Specification for Industrial Wire Cloth and Screens (Square Opening Series)

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F 593 - Standard Specification for Stainless Steel Bolts, Hex
Cap Screws, and Studs.
F 594 - Standard Specification for Stainless Steel Nuts.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INSTRUMENT SOCIETY OF AMERICA (ISA)

S75.02 - Control Valve Capacity Test Procedure

(Application for copies should be addressed to the Instrument Society of America, 67 Alexander Dr., P.O. Box 12277, Research Triangle Park, NC 27709.)

(Non-Government standards and other publications are normally avail-able from the organizations that prepare or distribute the documents. These documents also may be available in or through informational service.)

7.2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

7.3 Technical documents. Unless otherwise specified a technical manual and technical repair standard should be provided.

7.3.1 Technical content. The technical content of these manuals should include sufficient technical details to enable a repair, maintenance, or overhaul activity to restore the equipment dimensions, clearances, and tolerances such that the equipment is capable of performing its function as originally specified and is capable of being logistically supported by the DoD logistics support system.

7.3.2 Additional training aids. Unless otherwise specified (see 7.1), training aids such as audio or video tapes which supplements normal operations, trouble shooting and assembly/disassembly procedures of the unit should be provided. Additional documents or training aids must be acquired under a separate contract line item in the contract.

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7.4 First article. When a first article inspection is required, the items shall be a first article sample. The first article shall consist of the units specified. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the require-ment for samples for first article inspection to those bidders offering a product which has been previously acquired and tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

7.5 Sub-contracted material and parts. The packaging or delivery preparation requirements of referenced documents listed in section 7.2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

7.6 Ordering spare or repair parts. When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

7.6.1 Repair parts and repair parts boxes. Complete sets of repair basket elements for each type and class shall be furnished as follows:

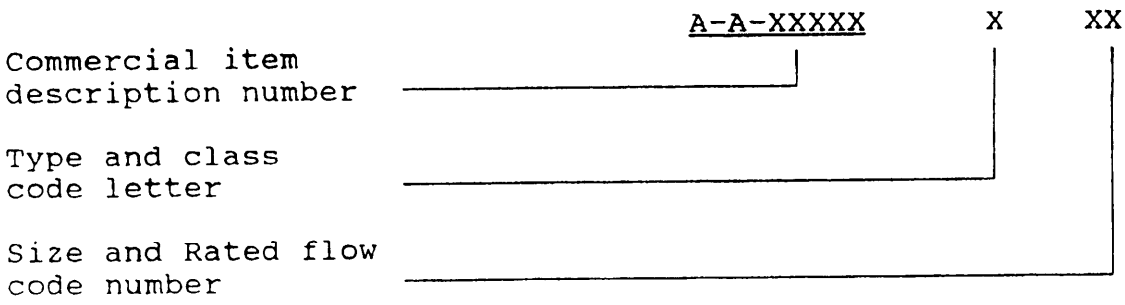
(a) One set for each of the ten initial strainers required.
(b) One set for each two additional strainers required.
(c) Cover gaskets - four for every strainer supplied.
(d) Other gaskets, seals and springs - one set for every ten strainers supplied.

When specified (see 7.1) items (c) and (d) shall be furnished in repair boxes.

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7.7 Preinstallation instruction. A set of instructions covering the preinstallation of the equipment should be furnished. Instructions should include all information necessary to return the unit to active status, such as, but not limited to: the addition of lubricants prior to operation, flushing of lines, removal of greaseproof barrier and the location of detached components. Instructions should be packaged in a transparent waterproof plastic bag, minimum 4 mil thick. Closure should be by heat sealing. The shipping container in which the instructions are packed should be so marked.

7.8 Part number. The part number is a definitive part number which corresponds to the type, class, size, and rated flow of strainers covered by this commercial item description. The commercial item description number, the type and class code letter, and the size and rated flow code numbers are combined to form a definitive part number. Part numbers for the strainers are assigned as follows:



7.8.1 Type and class. The type and class of strainer (see 2.1) are identified by a single letter (see Table VI).

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Table VI. Code letter to type and class.

CLASS	Type		
	I	II	III
1	A	B	C
2	D	E	F
3	G	H	I

7.8.2 Size and rated flow. The size and rated flow of strainer (see 2.1) are identified by a double digit number (see Table VII).

Table VII. Code number to size and rated flow.

	Rated flow (gpm)								
Size	75	150	300	450	600	750	900	1050	1200
2	01	-	-	-	-	-	-	-	-
3	-	02	-	-	-	-	-	-	-
4	-	-	03	-	-	-	-	-	-
6	-	-	-	04	05	06	-	-	-
8	-	-	-	-	-	-	07	08	09

7.9 Subject term (key word) listing.

Continuous
Screen
Cleaning
Filtration
Fittings
Motor
Fuel
Lube oil
Valves

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MILITARY INTERESTS:

Custodians:

Navy - SH
Air Force - 16

Preparing activity:
Navy - SH
(Project 4730-N106)

Review activities:
Army - AT, CR, ME
Navy - OS
Air Force - 10
DLA - CS

User activities:
Army - MR
Navy - CG, MC
Air Force - 90

Civil agency coordinating activities: (where appropriate)
AGR - APS
HHS - FEC

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